Course notes

In all this chapter, the coordinates are given in an orthonormal coordinate system of space. hypothesis is necessary to identify a normal vector from the coefficients of an equation and to calculate distances.

Adjusted lines

The right through $A(x_A, y_A, z_A)$ of a non-zero master vector (a, b, c) admits representation

$$d : \begin{cases} x = x_A + at \\ y = y_A + bt \\ z = z_A + ct \end{cases} \quad (t \in \mathbb{R}).$$

A point belongs to the right if it exists the same real t a right admits of many performances: change of starting point or multiplication of the director by a real non-zero.

Example. $(x, y, z) = (1, 2, 0) + t(2, -1, 3)$ describes the same line as $(3, 1, 3) + s(4, -2, 6)$; the link is $t = 1 + 2s$.

Cartesian plans

Equation of a plan

If $(a, b, c) \neq (0, 0, 0)$, the whole $ax + by + cz + d = 0$ is a normal vector plane (a, b, c). The normal plan $\vec{n}$ passing through A is built by writing $\overrightarrow{AM} \cdot \vec{n} = 0$.

Demonstration. The condition $\overrightarrow{AM} \cdot \vec{n} = 0$ written $a(x - x_A) + b(y - y_A) + c(z - z_A) = 0$. By developing, $d = -ax_A - by_A - cz_A$. Conversely, any satisfactory point this equation verifies the condition of orthogonality, therefore belongs to the plan.

Example. The normal plan $(2, -1, 3)$ passing through $A(1, 2, 0)$ is 2 $(x - 1) - (y - 2) + 3z = 0$, either $2x - y + 3z = 0$.

Intersections

For a right and a plan, replace x, y, z in the plane equation by their expressions parametric. A solution in t gives a point intersection; an identity gives a right included; a contradiction gives a disjointed parallel line. For two planes, normals do not colinears give an intersection on a straight line; solve the system with two equations in Choosing a variable as a parameter. If normals are collinear, the planes are parallel: Distinguished and disjointed. For two straight lines, use parameters s, t independent and solve the three equations. Without solution and with non-collinear directors, the straight ones are not coplanar.

Projection on a plan

If $\mathcal{P} : ax + by + cz + d = 0$, look for the planned M on the normal right passing through M : $H = M + t(a, b, c)$. Replace in the plan equation to determine t, then calculate MH. - We're in. deduced, without making it a formula to be learnt in isolation,

$$d(M, \mathcal{P}) = \frac{|ax_M + by_M + cz_M + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Verification is essential. After calculating a projected, check simultaneously $H \in \mathcal{P}$ and the colinearity of $\overrightarrow{MH}$ with normal.

Moving from objects to equations

12.1 One straight line, several settings

First use: S65.

Definition · Set a straight line

A line passing through A and a non-zero steering vector $\vec{u}$ is all M points such as that $\overrightarrow{AM} = t\vec{u}$, $t \in \mathbb{R}$. The parameter is neither a co-ordinated nor a time, unless the context specifies it.

Example resolved · Same set, other parameters

Expressions $(1+t;2-t;2t)$ and $(3+2s;-2s;4+4s)$ describe the same line: pose $t = 2+2s$ transforms the first into the second, and this transformation travels through all the real. The starting points may be different, as may proportional directors.

A1 · Immediate application Show that $(t;1+2t;3-t)$ and $(2+3s;5+6s;1-3s)$ describe the the same right. $(1;3;2)$ Does he belong there?

12.2 A normal fixed plane equation

First use: S67.

Property · Cartesian Equation

In an orthonormal coordinate system, a normal non-zero plane $(a;b;c)$ passing through $A = (x_A;y_A;z_A)$ a for equation $a(x-x_A)+b(y-y_A)+c(z-z_A) = 0$. It translates $\overrightarrow{AM} \cdot \vec{n} = 0$.

Example solved · Build and then test

For A=(1;0;2) and $\vec{n} = (2;-1;3)$, the plan has for equation 2 $x - y + 3z - 8 = 0$. Item (0;1;3) belongs to the plan because $0 - 1 + 9 - 8 = 0$. A point is not in the plan because its triplet looks like the normal vector.

A2 · Partial guidance Determine the normal plan $(1;2;-1)$ Test membership through (2.0;1). of (1;1;2), then give two non-collinear directions of the plan.

12.3 Three exits for one equation

First use: S69.

After substitution	Interpretation	Example
$at + b = 0,\ a \neq 0$	one point	straight stinging
$0t + b = 0,\ b \neq 0$	no point	strict parallel
$0t + 0 = 0$	all points	right included

Example Solved · Read a reduced equation

For the plane x+y+z=1 and the right $(t;2t;1-t)$, substitution gives 2 $t+1=1$, so t=0 and a single intersection (0;0;1). It is concluded only after reinjecting the value in the three coordinates.

A3 · Compare In the plan $z=1$, to study the positions of the right $d_1:(t;0;t)$, $d_2:(t;0;0)$ and $d_3:(t;0;1)$. Give the intersection point when it is unique and outline the three situations.

12.4 Two straight lines, two independent parameters

First use: S70.

Solved Example · Do not impose false simultaneity

For $d:(t;0;0)$ and $d':(1;1+s;0)$, the intersection requires t=1 and s=1. Imposing t=s would disappear Wrongly the point (1.0;0). The parameters are used separately to navigate each right.

A4 · Repair a Method For $d:(t;t;0)$ and $d':(1+s;1-s;s)$, solve with two parameters, then explain why the three coordinates should be checked.

12.5. Two verifications after the system

First use: S70.

Example Solved · Project on a Plan

Let $\mathcal{P}:x+2y-z=1$ and M=(1;1;1). Normal per M is $H=(1+t;1+2t;1-t)$. Replace in the plan gives 2 $+6t=1$, therefore $t=-1/6$. H=(5/6;2/3;7/6). $/6+4/3-7/6=1$ and $\overrightarrow{MH}=(-1/6)(1;2;-1)$. Distance is $\overline{6}/6$.

A5 · Synthesis with degenerate cases For a real m, d_m passes through (0;0;m) and has for direction $(1;m;1)$. Studying the intersection with $x+2y-z=1$. Distinguish m=0. Is there m for which the right is perpendicular to the plane?

A first step: Replace the three coordinates before dividing. For orthogonality, compare the director to A normal one.

A6 · Choice of strategy Determine the intersection of the planes $x+y+z=3$ and $x-y=0$. Provide one point and a guiding vector, then control these two informations in the equations.

Wait a minute at the courthouse. State the two properties that allow to recognize a project without indicate coordinates.

Review postponed · S71 and then S87 Match each line of an algebraic calculation to a geometric property; treat first the parameters that cancel a denominator.

Research activity

Three straight lines in front of a plan · 30 min.

In an orthonormal space coordinate system, the plane is considered $\mathcal{P}$ (d) equation $x+y+z=3$. In each question, the parameter t travel $\mathbb{R}$.

1. The right d_1 admits representation $(x;y;z)=(t;t;t)$. Determine $d_1\cap\mathcal{P}$.

2. The Right d_2 admits representation $(x;y;z) = (3+t;-t;0)$. Determine $d_2 \cap \mathcal{P}$.
3. The right d_3 admits representation $(x;y;z) = (t;-t;0)$. Determine $d_3 \cap \mathcal{P}$.
4. For each line, specify whether the equation obtained after substitution in the plane has a solution unique, no solution or any real solution. Linking these cases to geometric positions found.